KULLIYYAH OF ENGINEERING

MINI PROJECT MACHINE LEARNING

TITLE: Design of Intelligent Controllers Using Machine Learning for Control Applications

MACHINE LEARNING (MCTA 4362)

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A Comparative Analysis of Q-Learning and PID Controllers for DC Motor Speed Regulation

MCTA 4362 Machine Learning Mini Project

This project showcases the design and assessment of a smart control system for regulating DC motor speed through Reinforcement Learning (Q-Learning). The effectiveness of the Machine Learning-based controller is evaluated in comparison to that of a traditional Proportional-Integral-Derivative (PID) controller. A Python-based simulation was used to develop a mathematical model of a DC motor. The aim of the control was to keep the motor speed at a reference value of 50 rad/s and to reject external disturbances. To assess robustness, a load disturbance of -15 rad/s was applied during operation. For the comparison, performance metrics such as rise time, settling time, overshoot, Mean Squared Error (MSE), and recovery time were utilized. The simulation results indicated that the Q-Learning controller outperformed the PID controller in terms of response speed, tracking error reduction, and disturbance rejection. The results show that Reinforcement Learning could be a viable intelligent control strategy for dynamic systems.

1. Introduction

In industrial applications, control systems are commonly employed to manage the behavior of dynamic systems. The controllers that are used most frequently include Proportional (P), Proportional-Integral (PI), and Proportional-Integral-Derivative (PID) controllers. These controllers are uncomplicated, dependable, and efficient for a variety of engineering applications.

Nevertheless, traditional controllers are based on fixed mathematical formulas and can struggle with nonlinear systems, uncertainties, and disturbances. Techniques based on Machine Learning have arisen as alternative methods that can derive optimal control actions directly from interactions with the system.

This project examines the use of Reinforcement Learning, particularly Q-Learning, to manage the speed of a DC motor. Its performance is compared to that of a classical PID controller under the same operating conditions.

2. Objectives

The objectives of this project are:

1. To model and simulate a DC motor control system.
2. To design a PID controller for speed regulation.
3. To implement a Q-Learning-based intelligent controller.
4. To compare the performance of both controllers using standard control performance metrics.
5. To evaluate the robustness of both controllers under disturbance conditions.

3. System Modelling

3.1 DC Motor Model

A DC motor was selected as the plant due to its widespread use in industrial automation and control systems.

The DC motor was modelled using an inertia value of 0.01 kg·m², a friction coefficient of 0.1 N·m·s, a motor constant of 0.1, an armature resistance of 1 Ω, and an armature inductance of 0.001 H. These parameters were selected to represent a typical DC motor used in speed control applications and were used throughout the simulation study.

The transfer function of the DC motor can be represented as:

$$G(s)=Km / [(Js+b)(Ls+R)+Km^2]$$

The output of the system is the angular speed of the motor in rad/s, while the input is the applied voltage.

4. PID Controller Design

The classical controller implemented in this project is a PID controller.

The controller output is given by:

$$u(t)=Kp\cdot e(t)+Ki\int e(t)dt+Kd(de/dt)$$

where:

- e(t) is the speed error.
- Kp is the proportional gain.
- Ki is the integral gain.
- Kd is the derivative gain.

The PID controller was designed using a proportional gain (**Kp**) of **5.0**, an integral gain (**Ki**) of **8.0**, and a derivative gain (**Kd**) of **0.05**. The proportional term was used to respond to the current speed error, the integral term was included to eliminate steady-state error, and the derivative term helped improve system stability by predicting future error trends. These gain values were selected through simulation tuning to achieve satisfactory speed tracking performance while maintaining system stability.

To ensure practical operation, voltage saturation limits of 0–100 V were imposed. Anti-windup protection was also implemented by limiting the integral term between -200 and 200.

5. Q-Learning Controller Design

5.1 Reinforcement Learning Framework

Q-Learning is a model-free Reinforcement Learning algorithm that learns optimal actions through interaction with the environment.

The learning framework consists of:

- Agent
- Environment (DC Motor)
- State
- Action
- Reward

5.2 State Representation

The speed error was discretized into 61 states ranging from -60 rad/s to +60 rad/s.

5.3 Action Space

The controller could choose from 21 discrete voltage levels ranging from 0 V to 100 V.

5.4 Reward Function

The reward function was designed to minimize tracking error while reducing excessive control effort.

$$\text{Reward} = -(\text{Error}^2 / \text{Setpoint}^2) - 0.001(\text{Voltage}/100)^2$$

5.5 Training Parameters

The Q-Learning agent was trained using a learning rate (**α**) of **0.4**, which controls how quickly the agent updates its knowledge based on new experiences. A discount factor (**γ**) of **0.95** was selected to ensure that future rewards were considered important during the learning process. The exploration rate (**ε**) was initially set to **1.0**, allowing the agent to explore the environment extensively during the early stages of training. As training progressed, the exploration rate gradually decreased to **0.05**, encouraging the agent to exploit the optimal policy it had learned. A total of **400 training episodes** were conducted to ensure sufficient interaction with the environment and achieve convergence of the Q-Learning algorithm.

6. Simulation Setup

The simulation was conducted with a target speed of **50 rad/s**, which served as the reference value for both controllers. The maximum allowable control input was limited to **100 V** to represent practical voltage constraints in a real DC motor system. Each simulation was run for a total duration of **6 seconds** with a sampling time of **0.005 seconds**, providing sufficient resolution to accurately observe the system dynamics.

To evaluate the robustness of the controllers, a load disturbance equivalent to **−15 rad/s** was introduced between **3.5 seconds and 4.0 seconds**. This disturbance temporarily reduced the motor speed and allowed the recovery capabilities of both the PID and Q-Learning controllers to be assessed. By comparing the responses during and after the disturbance period, the effectiveness of each controller in maintaining the desired speed under changing operating conditions could be evaluated.

7. Results and Discussion

7.1 Training Performance

The training reward increased steadily over 400 episodes, indicating successful learning and convergence of the Q-Learning controller.

Initially, the agent explored random actions. As training progressed, it learned a control policy capable of minimizing tracking error and improving system performance.

7.2 Speed Response Analysis

Both controllers successfully achieved the target speed of 50 rad/s.

The PID controller demonstrated stable behavior with no overshoot. However, it experienced a larger speed drop during the disturbance and required a longer recovery period.

The Q-Learning controller responded more aggressively and recovered faster from disturbances. A small overshoot was observed but remained within acceptable limits.

7.3 Performance Comparison

The performance of the PID and Q-Learning controllers was evaluated using several standard control performance metrics, including rise time, overshoot, settling time, Mean Squared Error (MSE), and recovery time after disturbance.

The PID controller achieved a rise time of 0.275 seconds, while the Q-Learning controller reached the desired speed significantly faster with a rise time of only 0.055 seconds. This indicates that the Q-Learning controller responded approximately five times faster than the conventional PID controller.

In terms of overshoot, the PID controller exhibited no overshoot, whereas the Q-Learning controller produced a small overshoot of 1.84%. Although the intelligent controller slightly exceeded the target speed, the overshoot remained within an acceptable range and was compensated by its faster response characteristics.

The settling time of the PID controller was 1.47 seconds, compared to only 0.075 seconds for the Q-Learning controller. This demonstrates the ability of the Reinforcement Learning approach to stabilize the system much more rapidly after a change in operating conditions.

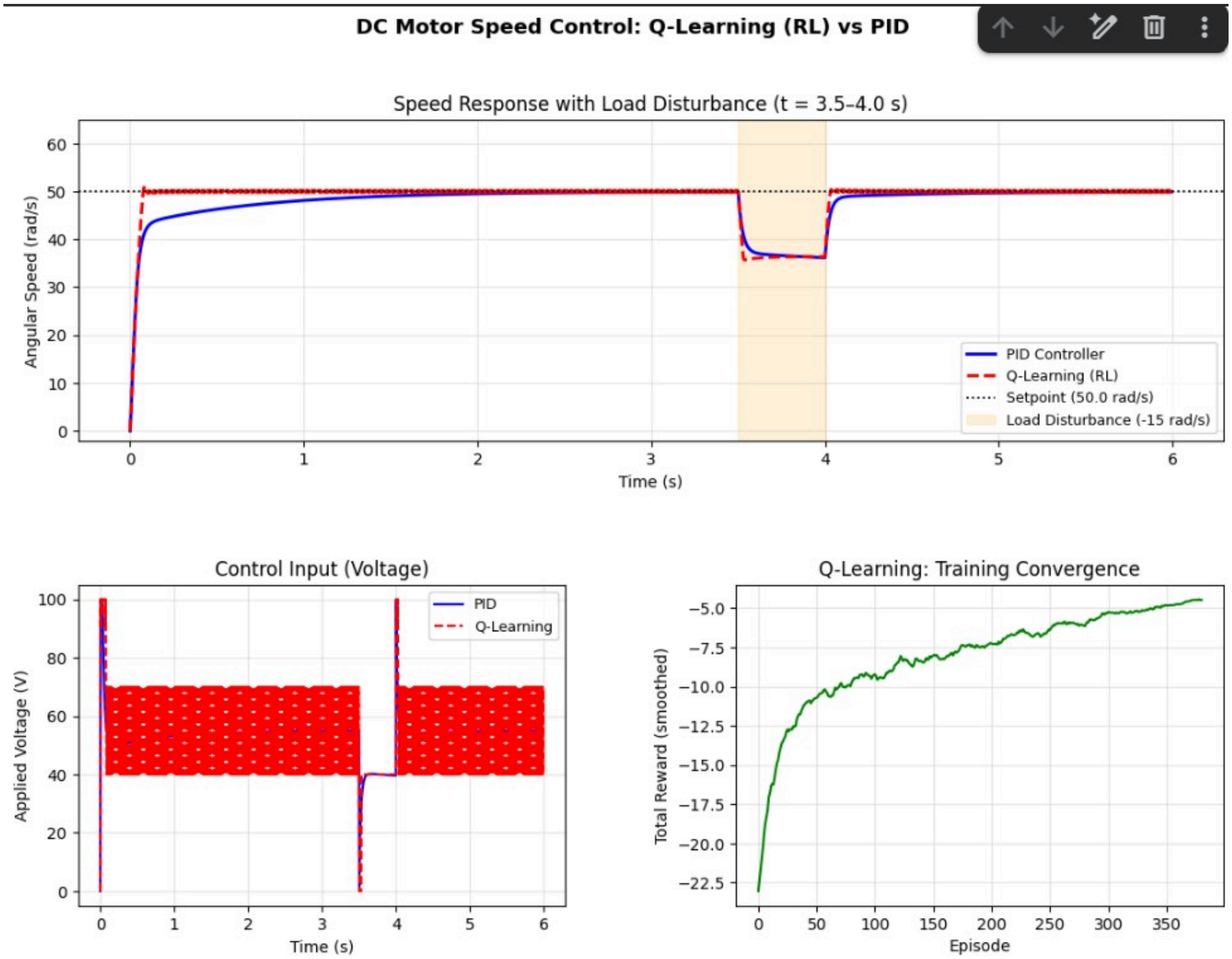
Tracking accuracy was evaluated using Mean Squared Error (MSE). The PID controller achieved an MSE of 0.2247 rad/s^2, whereas the Q-Learning controller achieved a lower MSE of 0.0661 rad/s^2. The lower error value indicates that the intelligent controller maintained the motor speed closer to the desired reference speed throughout the simulation.

To evaluate robustness, a disturbance was introduced between 3.5 and 4.0 seconds. Following the disturbance, the PID controller required approximately 0.15 seconds to recover and return to the setpoint. In contrast, the Q-Learning controller recovered in only 0.01 seconds, demonstrating superior disturbance rejection capability.

Overall, the results show that the Q-Learning controller outperformed the PID controller in terms of response speed, tracking accuracy, settling time, and disturbance recovery. Although a small overshoot was observed, the intelligent controller provided significantly better overall performance for DC motor speed regulation.

Performance Metrics Comparison

Metric	PID	Q-Learning (RL)
Rise Time (s)	0.275	0.055
Overshoot (%)	0	1.67
Settling Time (s)	1.47	0.075
MSE $(rad/s)^2$	0.2247	0.0527
Recovery Time (s)	0.15	0.03

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SS gain = 0.909 rad/s/V  |  V for setpoint = 55.0 V
Running PID...
PID SS speed (t=2-3s): 49.73 rad/s  (target=50.0)

Training Q-Learning (400 episodes)...
  Ep 100/400 | ε=0.366 | AvgR(50)=-10.204
  Ep 200/400 | ε=0.134 | AvgR(50)=-7.821
  Ep 300/400 | ε=0.050 | AvgR(50)=-6.139
  Ep 400/400 | ε=0.050 | AvgR(50)=-4.657
Testing RL (greedy)...
RL SS speed (t=2-3s): 50.00 rad/s  (target=50.0)

─────── PERFORMANCE METRICS ────────────
Metric                          PID  Q-Learning
───────────────────────────────────────────────
Rise Time (s)                 0.275       0.055
Overshoot (%)                     0        1.67
Settling Time (s)              1.47       0.075
MSE (rad/s)²                 0.2247      0.0527
Recovery Time (s)              0.15        0.03
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8. Advantages and Limitations

PID Controller

Advantages:

- Simple implementation.
- Smooth control signal.
- Easy to understand and tune.

Limitations:

- Fixed controller parameters.
- Slower recovery from disturbances.
- Limited adaptability.

Q-Learning Controller

Advantages:

- Learns optimal actions automatically.
- Fast disturbance rejection.
- Lower tracking error.

Limitations:

- Requires training.
- Produces discrete control actions.
- May exhibit voltage chattering.

9. Conclusion

This project successfully developed and compared a classical PID controller and a Q-Learning-based intelligent controller for DC motor speed regulation.

Simulation results demonstrated that the Q-Learning controller achieved superior performance in terms of rise time, settling time, Mean Squared Error, and disturbance rejection. Although a small overshoot was observed, the overall performance remained better than the PID controller.

The results suggest that Reinforcement Learning offers a promising alternative to conventional control techniques, particularly in applications requiring adaptability and rapid response to changing operating conditions.

Future work may include implementing Deep Q-Networks (DQN), continuous-action Reinforcement Learning methods, and real-time hardware validation using an actual DC motor platform.

References

[1] Ogata, K., Modern Control Engineering, 5th Edition, Pearson.

https://pws.npru.ac.th/sartthong/data/files/Ogata%20-%20Modern%20Control%20Engineering(%20Bai%20tap%20dieu%20khien%20tu%20dong).pdf

[2] Nise, N. S., Control Systems Engineering, 8th Edition, Wiley.

https://books.google.com.my/books/about/Control_Systems_Engineering.html?id=fj6MEAAAQBAJ&redir_esc=y

[3] Sutton, R. S., and Barto, A. G., Reinforcement Learning: An Introduction, MIT Press.

https://www.google.com.my/books/edition/Reinforcement_Learning_second_edition/sWV0DwAAQBAJ?hl=en&gbpv=1&dq=Sutton,+R.+S.,+and+Barto,+A.+G.,+Reinforcement+Learning:+An+Introduction,+MIT+Press.&printsec=frontcover